

Characterisation and Calibration of the DADOS Long-Slit Spectrograph on the PlaneWave CDK17 Telescope

Hanaan Khan

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Abstract

The PlaneWave CDK17 telescope on top of the Tolansky Building at Royal Holloway, University of London has recently been equipped with a Baader Planetarium DADOS long-slit spectrograph, providing a new mode of astronomical observation. This project presents a systematic calibration of the instrument using a Python pipeline developed for this purpose. The complementary metal-oxide-semiconductor (CMOS) detector was characterised across temperatures from -20 to $+10$ °C, yielding a bias level of 240.1 ± 0.2 analogue-to-digital units (ADU), a read noise of ~ 0.37 ADU, and a dark current rate of 0.0256 ADU s^{-1} pixel^{-1} at -17 °C. A hot corner region showing dark current $8.35\times$ above the detector mean was identified and attributed to a thermal gradient from the thermoelectric cooler. No suitable flat field calibration was available in the current dataset: the slit-viewing camera (C0) does not sample the spectrograph optical path, and the LED illumination source was found to be non-uniform. A wavelength solution covering $6143\text{--}7174$ Å was derived from a neon arc lamp, giving a dispersion of 1.346 Å pixel^{-1} and a resolving power of $R \approx 1520$. The results are presented as a preliminary instrument manual to support future nighttime observations.

1 Introduction

Astronomical spectrographs are among the most powerful instruments available to observational astronomers, allowing light from celestial objects to be dispersed into its constituent wavelengths. The resulting spectra contain a wealth of physical information, including the chemical composition, temperature, radial velocity and redshift of the source [1]. Before any scientific observations can be made, a spectrograph must be carefully calibrated to ensure that the data it produces are reliable and quantitatively sound [2].

The University has recently commissioned a new long-slit spectrograph mounted on the PlaneWave CDK17 telescope on top of the Tolansky Building. Since the instrument is newly installed, a systematic calibration procedure must be established before it can be used for scientific observations. All data for this project were acquired during the daytime under controlled laboratory conditions.

Calibration of a CMOS-based spectrograph involves four main stages: bias subtraction to remove the fixed electronic offset present in every detector readout [3]; dark current subtraction to account for thermally generated charge that accumulates during an exposure [3]; flat field correction to remove the non-uniform spectral response of the instrument; and wavelength calibration, which maps pixel position to wavelength by matching observed emission lines from an arc lamp to their known laboratory values [2]. This project carries out each of these steps and characterises the performance of the detector across a range of operating temperatures, with the aim of producing a calibration protocol suitable for future nighttime observations. The report is structured as follows. Section 2 provides the theoretical background necessary to understand the calibration process,

including a description of charge-coupled device (CCD)/CMOS noise sources and the principles of spectral dispersion. Section 3 describes the telescope, spectrograph, and detector used in this project. Sections 4–7 present the calibration analysis for each frame type in turn. Section 8 summarises the results and discusses directions for future work.

2 Background

2.1 Spectral Dispersion and Grisms

Dispersion in a grating spectrograph is governed by the grating equation [4]:

$$m\lambda = d(\sin \alpha + \sin \beta), \quad (1)$$

where m is the (dimensionless) diffraction order, λ is the wavelength (in nm), d is the groove spacing (in mm), α is the angle of incidence (in radians), and β is the angle of diffraction (in radians). The spectrograph used here employs a *grism*, a diffraction grating cemented to a prism such that light at a central wavelength passes straight through without deflection [5]. The grism has a groove density of 200 lines mm^{-1} , giving wavelength coverage appropriate for low-resolution spectroscopy in the red optical regime. The spectral resolution is defined as:

$$R = \frac{\lambda}{\Delta\lambda}, \quad (2)$$

where λ and $\Delta\lambda$ are in the same units (typically Å or nm), making R dimensionless. $\Delta\lambda$ is measured from the full width at half maximum (FWHM) of a resolved arc lamp emission line, as described in Section 7.

2.2 CMOS Detectors and Noise Sources

Astronomical CMOS detectors convert incident photons into charge via the photoelectric effect, which is read out as a digital signal in analogue-to-digital units (ADU) [3]. Unlike CCDs, CMOS sensors use per-pixel amplifiers, allowing faster readout and lower read noise, though this can introduce fixed pattern noise [6]. Four noise sources are relevant to this project.

Bias: a fixed electronic offset added to every pixel readout to keep the signal positive before digitisation. It is measured from zero-second exposures and removed by subtracting a master bias, formed as the median of multiple bias frames [3]:

$$B_{\text{master}}(x, y) = \text{median}\{B_1(x, y), \dots, B_n(x, y)\}, \quad (3)$$

where x and y are pixel coordinates (dimensionless integers).

Dark current: thermally generated charge that accumulates during an exposure even with the shutter closed. Its rate follows an Arrhenius relationship with temperature [7]:

$$D \propto T^3 \exp\left(-\frac{E_g}{2k_B T}\right), \quad (4)$$

where T is the absolute temperature (in K), E_g is the semiconductor bandgap energy (in eV), and $k_B = 1.381 \times 10^{-23} \text{ J K}^{-1}$ is Boltzmann’s constant. The total dark signal scales linearly with exposure time, $D_{\text{total}} = D \cdot t$, so a dark frame taken at the same temperature and exposure time can be subtracted directly.

Hot pixels: isolated pixels with anomalously high dark current due to crystal lattice defects [3]. Their signal grows strongly with temperature and they must be masked before spectral extraction.

Read noise σ_r : a fixed noise floor introduced during analogue-to-digital conversion, independent of exposure time [3].

2.3 Flat Field and Wavelength Calibration

A flat field frame, taken by illuminating the spectrograph with a uniform source, captures the combined effect of pixel sensitivity variations, vignetting, and the wavelength-dependent throughput of the instrument. After bias and dark subtraction, the flat is normalised by its median to produce $F(x, y) \approx 1$, and science frames are corrected as [2]:

$$S_{\text{corrected}}(x, y) = \frac{S(x, y) - B(x, y) - D(x, y)}{F(x, y)}, \quad (5)$$

where all quantities are in ADU and $F(x, y)$ is dimensionless after normalisation.

A wavelength solution maps pixel position to wavelength by fitting a polynomial to matched pairs of pixel centres and known arc lamp wavelengths [8]. Pixel centres are determined by fitting a Gaussian to each emission line:

$$I(x) = A \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) + B, \quad (6)$$

where I is the intensity (in ADU), x is the pixel position (dimensionless), A is the peak amplitude (in ADU), μ is the line centre (in pixels), σ is the standard deviation (in pixels), and B is the background level (in ADU). The FWHM is related to σ by $\text{FWHM} = 2\sqrt{2\ln 2}\sigma \approx 2.355\sigma$ (in pixels). The resulting polynomial solution $\lambda(x) = \sum_k a_k x^k$ is assessed by the root mean square (RMS) of residuals between fitted and known wavelengths, and the dispersion in $\text{\AA pixel}^{-1}$ is its derivative evaluated at the centre of the calibrated range.

3 Instrumentation

3.1 Telescope

The observations for this project were carried out using the PlaneWave CDK17 telescope located in the observatory. The telescope has an aperture of 432 mm and a focal length of 2939 mm, giving a focal ratio of $f/6.8$.

3.2 Spectrograph

The spectrograph is a long-slit design, meaning that the entrance aperture is a narrow rectangular slit rather than a circular aperture. This configuration allows a one-dimensional spatial profile of the target to be recorded simultaneously with the spectral information, with the spatial direction oriented perpendicular to the dispersion direction on the detector [4]. The slit width determines the spectral resolution: a narrower slit admits less light but produces sharper spectral lines. It should be noted that the DADOS is designed for use with telescopes operating at $f/10$ or slower [9], whereas the PlaneWave CDK17 operates at $f/6.8$. The faster focal ratio means the beam entering the spectrograph is wider than the instrument was designed for, which can cause vignetting at the edges of the slit and may contribute to the $\sim 13\%$ variation in spectral response observed in the flat field (Section 6).

Dispersion is provided by a grism with a groove density of 200 lines mm^{-1} . As described in Section 2.1, a grism combines a diffraction grating with a prism such that light at the blaze wavelength passes straight through without deflection [5]. The 200 lines mm^{-1} groove density gives a dispersion of approximately $1.34 \text{ \AA pixel}^{-1}$ at the centre of the detector, as measured from the wavelength solution derived in Section 7. This places the instrument in the low-resolution regime, appropriate for characterising the broad spectral features of extended or faint sources.

A slit-viewing camera (the C0 camera) is mounted separately and is used to image the slit region directly, allowing the user to verify that a target is correctly placed on the slit before taking a spectrum. The slit camera operates at a different pixel scale to the spectrograph detector, which is why the flat field frames from the two cameras have different array dimensions, as discussed in Section 6.

3.3 Detector

The spectrograph is equipped with a Moravian C2-7000A CMOS camera, housing a Sony IMX428 global-shutter sensor. Unlike a traditional CCD, a CMOS sensor has per-pixel amplifiers, which allows faster readout and lower read noise, though it can introduce additional fixed-pattern noise [6]. The detector array has dimensions of 3216×2208 pixels, with a pixel size of $4.50 \times 4.50 \mu\text{m}$. The camera is thermoelectrically cooled, allowing the operating temperature to be set by the user. For this project, data were acquired at temperatures ranging from -20°C to $+10^\circ\text{C}$ in order to characterise the temperature dependence of the dark current, as described in Section 5.

All raw frames were stored as flexible image transport system (FITS) files [10], which is the standard format for astronomical imaging data. The calibration pipeline was written in Python, making use of the ASTROPY [11], NUMPY [12], and SCIPY [13] libraries.

3.4 Assumptions and Approximations

Several simplifying assumptions are made throughout this calibration analysis. The detector gain is assumed to be approximately $1 \text{ e}^- \text{ ADU}^{-1}$ [6], used only when converting ADU figures to electrons for comparison with literature values. The master bias is assumed to provide a valid correction at any temperature, justified by the thermal stability shown in Section 4. The dark current rate at -17°C is assumed to scale linearly with exposure time, verified in Section 5.2. No flat field calibration applicable to science spectra was available in the current dataset; the implications of this are discussed in Section 6.

4 Bias Correction

The bias level of the CMOS detector was characterised by acquiring between 5 and 10 bias frames at each of six detector temperatures: -20.0 , -18.2 , -17.0 , -10.0 , 0.0 , and $+10.0^\circ\text{C}$, giving 50 frames in total. At each temperature, the individual frames were median combined to reject cosmic ray hits and produce a master bias. The mean bias level, pixel-to-pixel standard deviation, and standard error on the mean at each temperature are summarised in Table 1.

4.1 Bias Level and Thermal Stability

The mean bias level across all temperatures is $240.1 \pm 0.2 \text{ ADU}$. The full range spans only 240.072 ADU at -20°C to 240.222 ADU at $+10^\circ\text{C}$, a variation of 0.15 ADU across a 30°C temperature range. As shown in Figure 1, the variation is smaller than the standard error at any individual temperature,

Table 1: Mean bias level, pixel-to-pixel standard deviation, and standard error on the mean at each detector temperature. The standard error is computed as $\sigma_{\bar{x}} = \sigma/\sqrt{n}$, where n is the number of frames combined.

Temperature ($^{\circ}\text{C}$)	n	Mean (ADU)	σ (ADU)	$\sigma_{\bar{x}}$ (ADU)
−20.0	5	240.072	0.514	0.230
−18.2	10	240.091	0.362	0.114
−17.0	5	240.129	0.528	0.236
−10.0	10	240.132	0.390	0.123
0.0	10	240.173	0.415	0.131
+10.0	10	240.222	0.446	0.141

confirming that the scatter is consistent with noise rather than a genuine thermal dependence. A single master bias taken at any convenient temperature is therefore sufficient to calibrate observations across the full operating range.

The pixel-to-pixel standard deviation is consistently 0.36–0.53 ADU across all temperatures, representing the combined contribution of read noise and fixed pattern noise. Assuming a typical CMOS gain of $\sim 1 \text{ e}^- \text{ ADU}^{-1}$, this corresponds to a read noise of $\sim 0.4 \text{ e}^- \text{ RMS}$, consistent with modern back illuminated CMOS sensors [6].

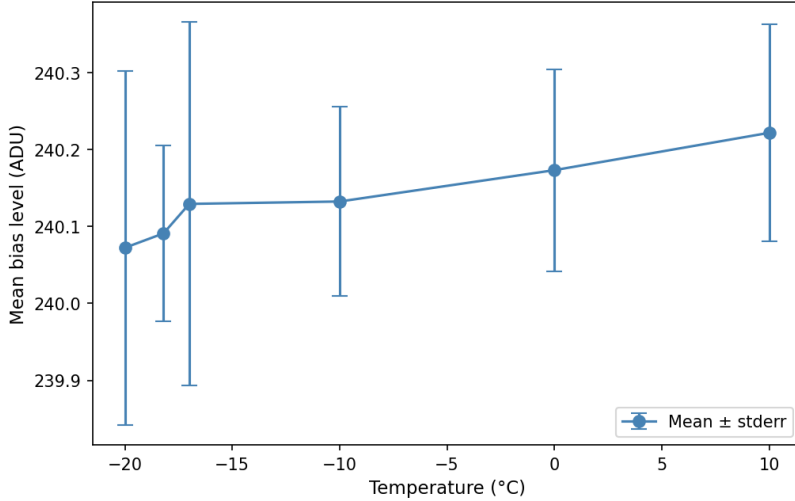


Figure 1: Mean bias level as a function of detector temperature. Error bars represent the standard error on the mean ($\sigma/\sqrt{n}$). The bias level is consistent at 240.1 ADU across all temperatures, with a total variation of 0.15 ADU over a 30°C range.

4.2 Spatial Structure and Fixed Pattern Noise

Figures 2–4 show the master bias frames at -20°C , -17°C , and $+10^{\circ}\text{C}$. At low temperatures the frames are spatially uniform, but at $+10^{\circ}\text{C}$ faint horizontal striping becomes visible, caused by slight offsets between the per-row readout amplifiers of the CMOS sensor [6]. Since this fixed pattern noise is stable and reproducible, it is captured in the master bias and removed by subtraction. The histograms in Figures 2–4 remain narrow and symmetric at all temperatures ($\sigma \sim 0.4 \text{ ADU}$), confirming that no residual pattern remains after combination. The temperature dependence of the striping suggests that amplifier offsets become more pronounced as the detector warms, providing an additional motivation for operating at the cooled nominal temperature.

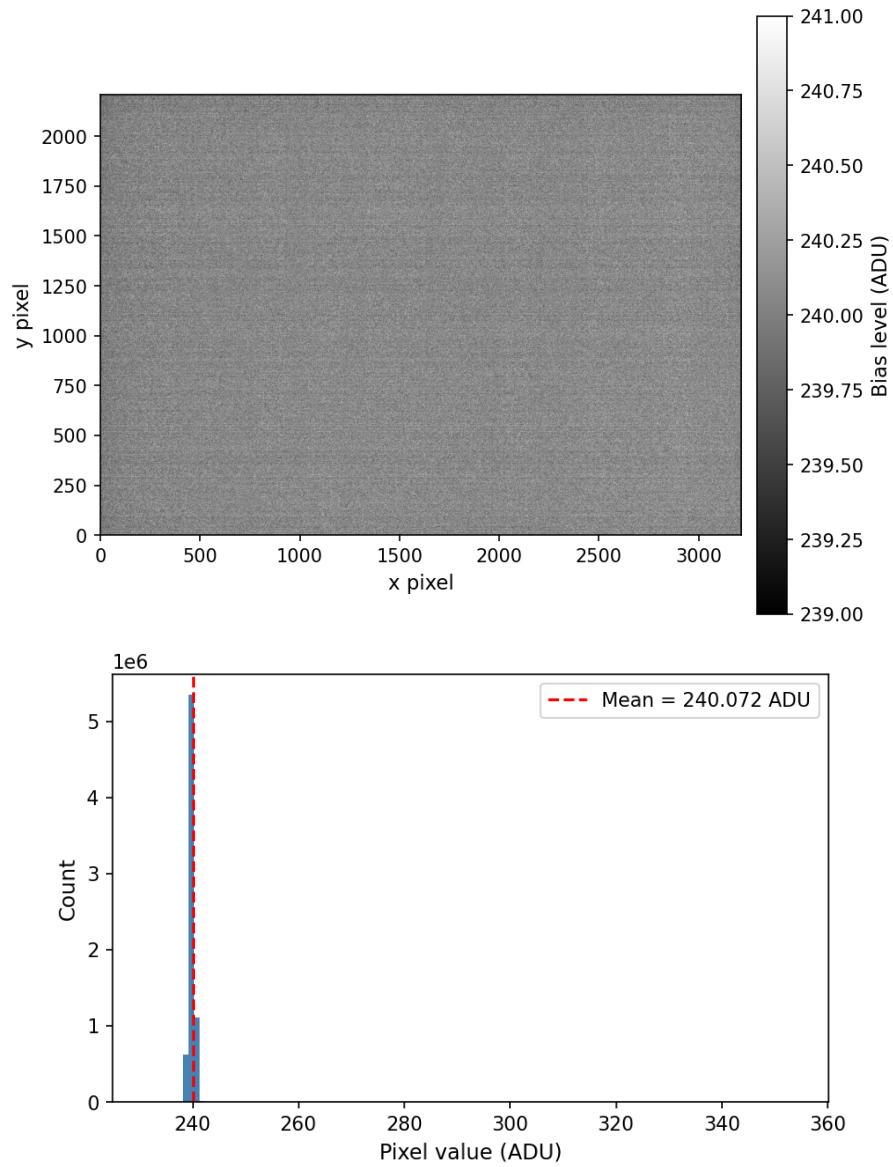


Figure 2: Master bias frame at -20°C (top) and corresponding pixel value distribution (bottom). The frame is spatially uniform at this temperature, with no visible fixed pattern noise.

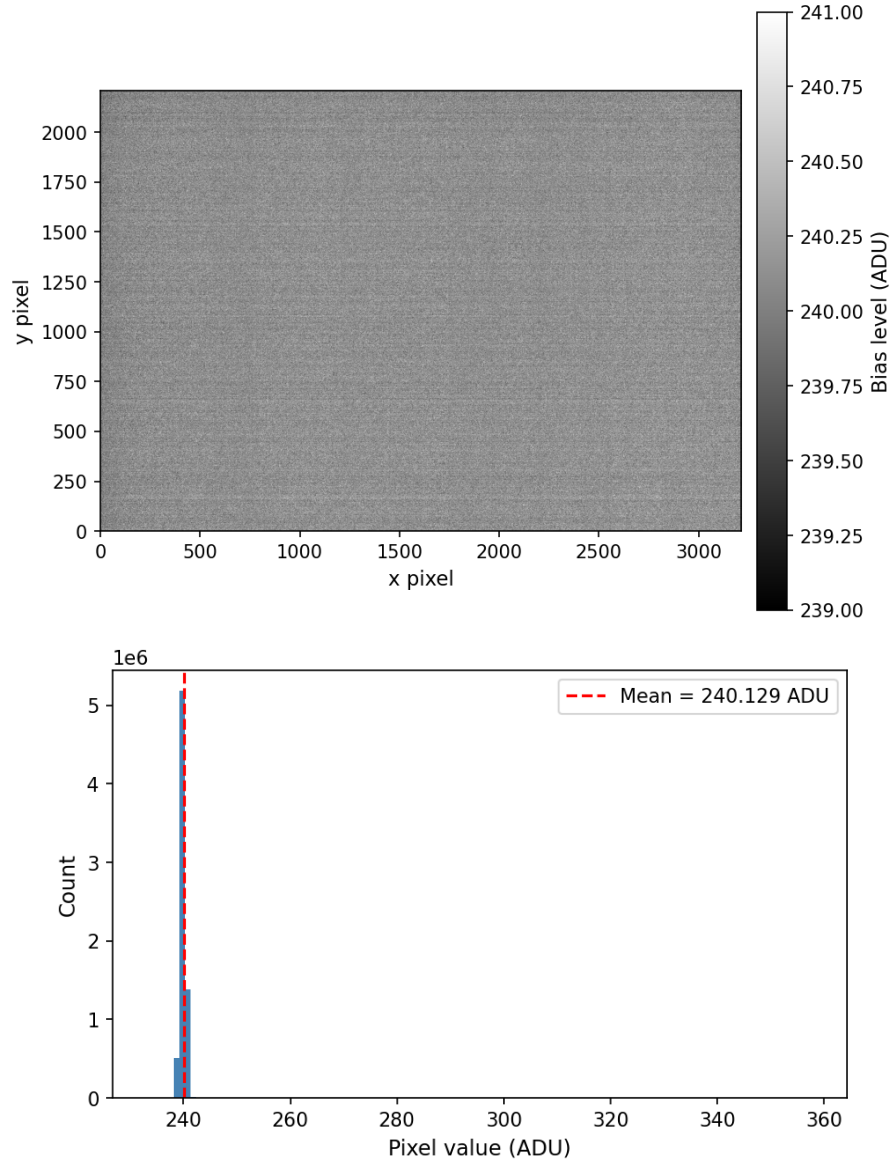


Figure 3: Master bias frame at -17°C (top) and corresponding pixel value distribution (bottom). The nominal operating temperature shows a similarly uniform frame with a narrow symmetric histogram ($\sigma \sim 0.4 \text{ ADU}$).

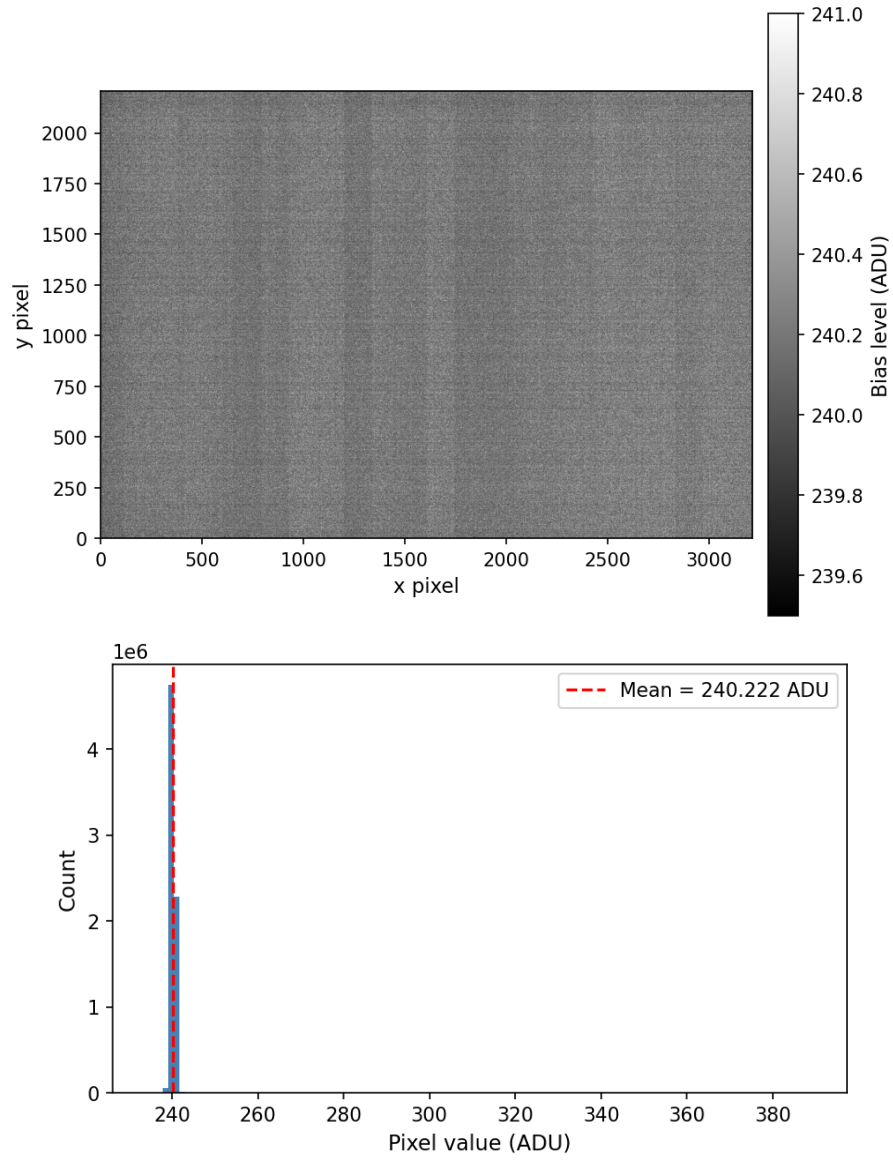


Figure 4: Master bias frame at $+10^{\circ}\text{C}$ (top) and corresponding pixel value distribution (bottom). Faint horizontal striping, indicative of CMOS fixed pattern noise from per-row amplifier offsets, becomes clearly visible at this elevated temperature.

4.3 Read Noise Estimate

A more rigorous read noise estimate is obtained from the pixel-by-pixel difference of two bias frames at the same temperature, which cancels the fixed pattern and isolates the random read noise [3]:

$$\sigma_{\text{read}} = \frac{\sigma_{\text{diff}}}{\sqrt{2}}, \quad (7)$$

where σ_{diff} is the standard deviation of the pixel-by-pixel difference frame, and σ_{read} is the per-pixel read noise, both in ADU. Using the -17°C frames, $\sigma_{\text{diff}} \approx 0.53$ ADU gives $\sigma_{\text{read}} \approx 0.37$ ADU per pixel, confirming the detector is well-suited to low-signal spectroscopic observations.

5 Dark Current

Dark frames were acquired at five detector temperatures (-20.0 , -17.0 , -10.0 , 0.0 , and $+10.0^\circ\text{C}$) and at exposure times ranging from 1 to 30 s, giving 45 frames in total across 9 temperature–exposure time groups. At each temperature, 5 frames per group were median combined after subtracting the nearest temperature master bias. The mean dark current, standard error, and dark current rate for all groups are summarised in Table 2.

Table 2: Mean dark current (bias subtracted), standard error on the mean, and dark current rate for each temperature–exposure time group. The dark current rate is derived from the linear fit to the -17°C multi-exposure data (see Section 5.2); single-point groups are marked with a dash. All values are given in ADU.

Temperature ($^\circ\text{C}$)	Exposure (s)	n	Mean (ADU)	$\sigma_{\bar{x}}$ (ADU)	Rate (ADU s^{-1})
-20.0	2	5	0.050	0.0013	—
-17.0	1	5	0.032	0.0017	0.0256
-17.0	2	5	0.054	0.0023	
-17.0	5	5	0.128	0.0018	
-17.0	10	5	0.254	0.0022	
-17.0	30	5	0.767	0.0013	
-10.0	30	5	0.816	0.0016	—
0.0	30	5	0.871	0.0022	—
$+10.0$	30	5	0.987	0.0022	—

5.1 Temperature Dependence

The mean dark current increases monotonically with detector temperature, consistent with the Arrhenius relationship in Equation 4. Considering the 30 s exposure frames, the dark current rises from 0.050 ADU at -20°C to 0.987 ADU at $+10^\circ\text{C}$, approximately $1.97\times$ that at -17°C . The absolute values remain below 1 ADU even at $+10^\circ\text{C}$, indicating excellent intrinsic dark current performance. For typical science exposures of 30–120 s at -17°C , the dark current contribution will be at most ~ 3 ADU, negligible for most observing programmes.

5.2 Linearity with Exposure Time

The -17°C group provides the only dataset with multiple exposure times (1, 2, 5, 10, and 30 s), enabling a direct test of linearity as predicted by $D_{\text{total}} = D \cdot t$. Figure 5 shows the mean dark current with a least-squares linear fit constrained to pass through the origin, giving a dark current rate of $D = 0.0256$ ADU s^{-1} pixel $^{-1}$. The measured values are consistent with predicted values of

0.026, 0.051, 0.128, 0.256, and 0.768 ADU at 1, 2, 5, 10, and 30 s respectively, with a maximum deviation from linearity of less than 0.003 ADU. This confirms that dark frame scaling is valid for this detector.

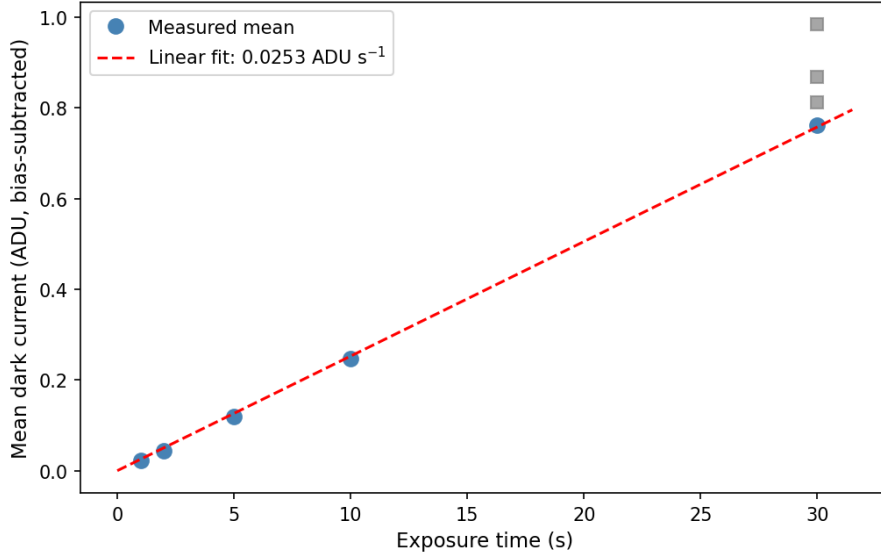


Figure 5: Mean dark current as a function of exposure time at -17°C , with a linear fit (dashed line) giving a dark current rate of $0.0256 \text{ ADU s}^{-1} \text{ pixel}^{-1}$. Single-point measurements at other temperatures are shown for comparison.

5.3 Hot Pixels

Hot pixels were identified in the -17°C , 30 s master dark frame as those with values exceeding 5σ above the mean dark current level. A total of 4338 hot pixels were found, representing approximately 0.06% of the total detector area of $3216 \times 2208 = 7,100,928$ pixels. Their spatial distribution is shown in Figure 6, and the five most prominent are listed in Table 3.

Table 3: The five most prominent hot pixels identified in the -17°C , 30 s master dark frame, with dark current values at -17°C and $+10^{\circ}\text{C}$ to illustrate the temperature dependence. Pixel coordinates are given as (x, y) in detector pixel space.

Pixel (x, y)	Dark current at -17°C (ADU)	Dark current at $+10^{\circ}\text{C}$ (ADU)	Increase factor
(3033, 4)	~ 27	~ 295	$\sim 10.9\times$
(3089, 0)	~ 1	~ 72	$\sim 72\times$
(3193, 0)	~ 1	~ 13	$\sim 13\times$
(3091, 9)	~ 6	~ 1	—
(1653, 14)	~ 2	~ 2	$\sim 1\times$

Pixel (3033, 4) is the most extreme case, rising from ~ 27 ADU at -17°C to ~ 295 ADU at $+10^{\circ}\text{C}$, an increase of $10.9\times$ over a 27°C range, consistent with the Arrhenius model [7]. All tracked hot pixels show near-zero dark current at -20°C , confirming that cooling effectively suppresses hot pixel contamination. Saturated hot pixels cannot be corrected by dark subtraction and must be masked before spectral extraction; the spatial map in Figure 6 provides the basis for constructing such a mask. The temperature dependence of the five most prominent hot pixels is shown in

Figure 7.

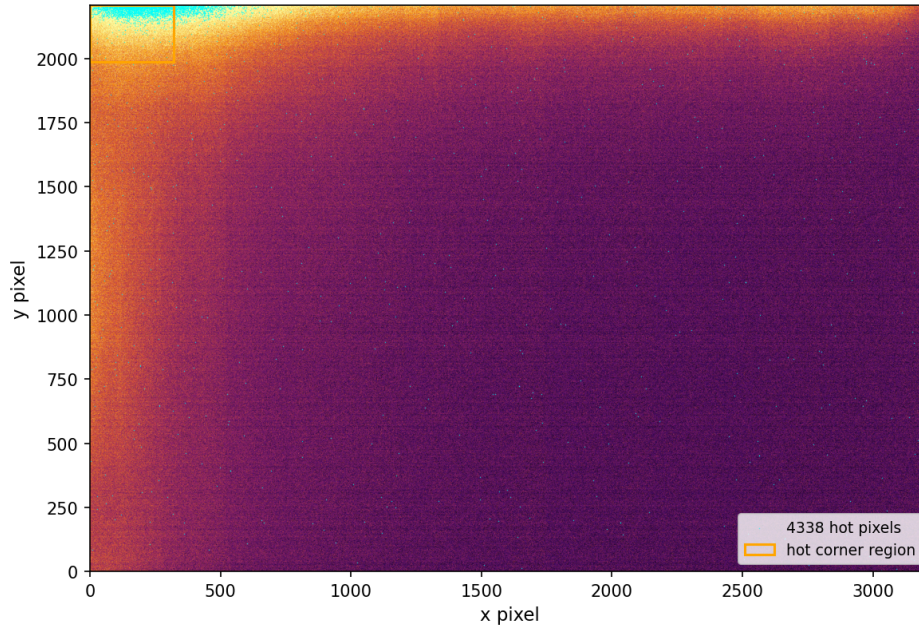


Figure 6: Spatial distribution of the 4338 hot pixels (cyan points) identified in the -17°C , 30 s master dark frame. The dense concentration of hot pixels in the upper-left corresponds to the hot corner region discussed in Section 5.4.

5.4 Hot Corner

In addition to isolated hot pixels, a spatially extended region of elevated dark current is present in the top-left corner of the detector. Figures 8–11 show the 30 s master dark frames at all four available temperatures on a shared colour scale. At -17°C the corner is only marginally brighter than the rest of the detector, but by $+10^{\circ}\text{C}$ a visibly brighter patch extends across approximately the top-left 220×321 pixels.

The mean dark current in the top-left 220×321 pixel region is 4.086 ADU, compared to 0.490 ADU for the remainder of the detector, a factor of $8.35\times$ higher. The most likely cause is a localised thermal gradient from the thermoelectric cooler geometry; even a local temperature excess of $1\text{--}2^{\circ}\text{C}$ would be sufficient to produce an $8\times$ increase in dark current given the exponential sensitivity of Equation 4 [7]. The hot corner region is also marked in Figure 6, where the dense concentration of hot pixels in the upper-left is clearly visible. Future users should avoid placing science targets in the top ~ 220 rows of the detector.

6 Flat Field Correction

A flat field calibration requires a diffuse, uniformly illuminated source that follows the same optical path as the science target: light must pass through the telescope aperture, enter the spectrograph slit, and traverse the full dispersing optical train before reaching the science detector [2]. Only then will the flat frame encode the combined wavelength-dependent throughput of the grism, the pixel-to-pixel sensitivity variations of the detector, and any dust or vignetting in the beam.

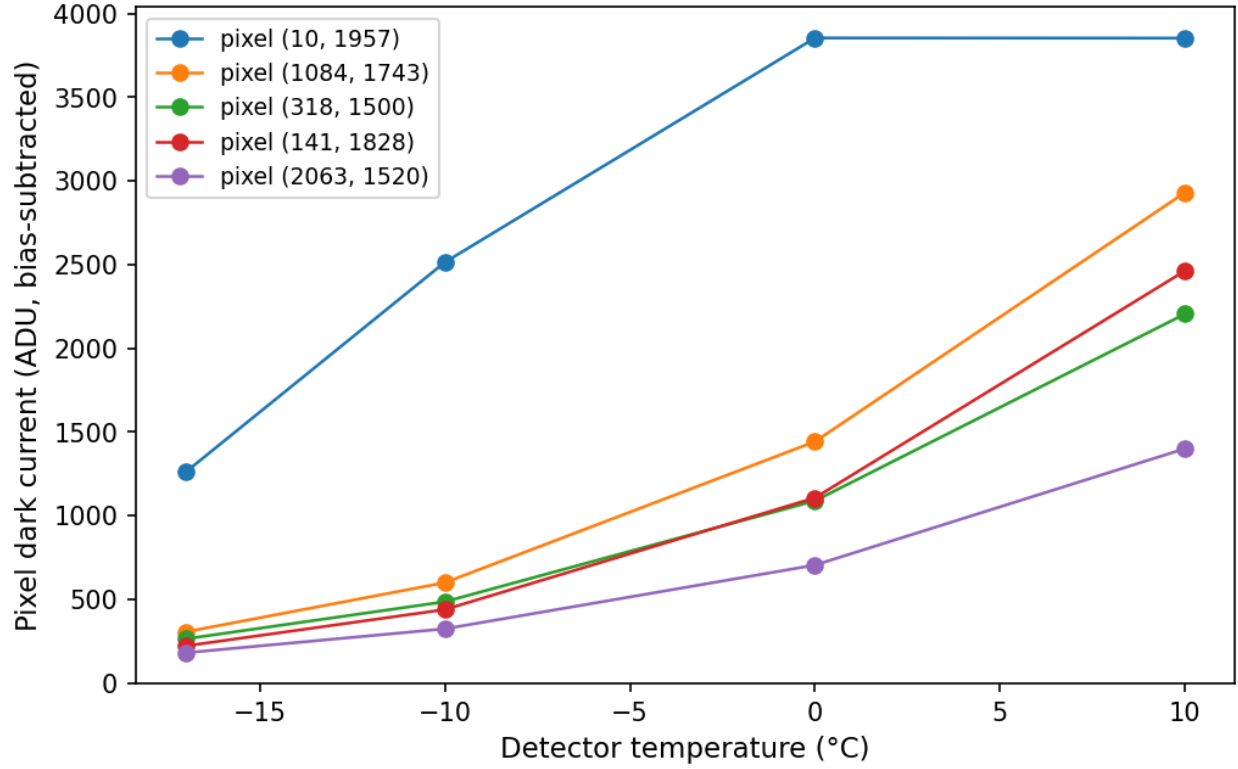


Figure 7: Dark current of the four most temperature-sensitive hot pixels as a function of detector temperature. Pixel (1653, 14) is omitted as its dark current is effectively constant at ~ 2 ADU across all temperatures, showing no meaningful thermal dependence. The strongly super-linear increase in the remaining four is consistent with the Arrhenius relationship (Equation 4). At -20°C all hot pixels are effectively suppressed.

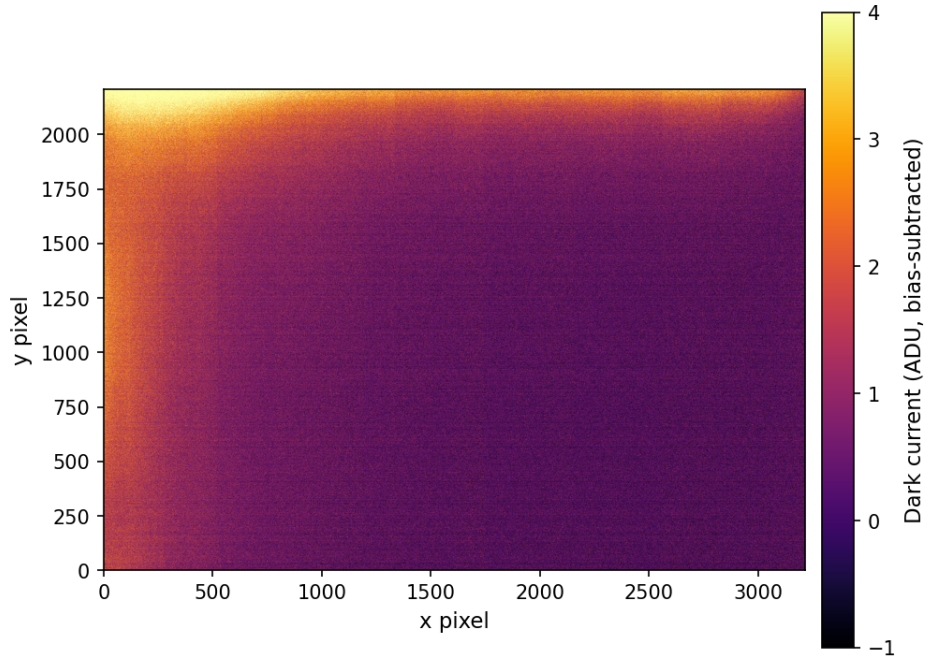


Figure 8: Master dark frame, 30 s exposure at -17°C . At the nominal operating temperature dark current is negligible and the hot corner is only marginally elevated above the detector mean.

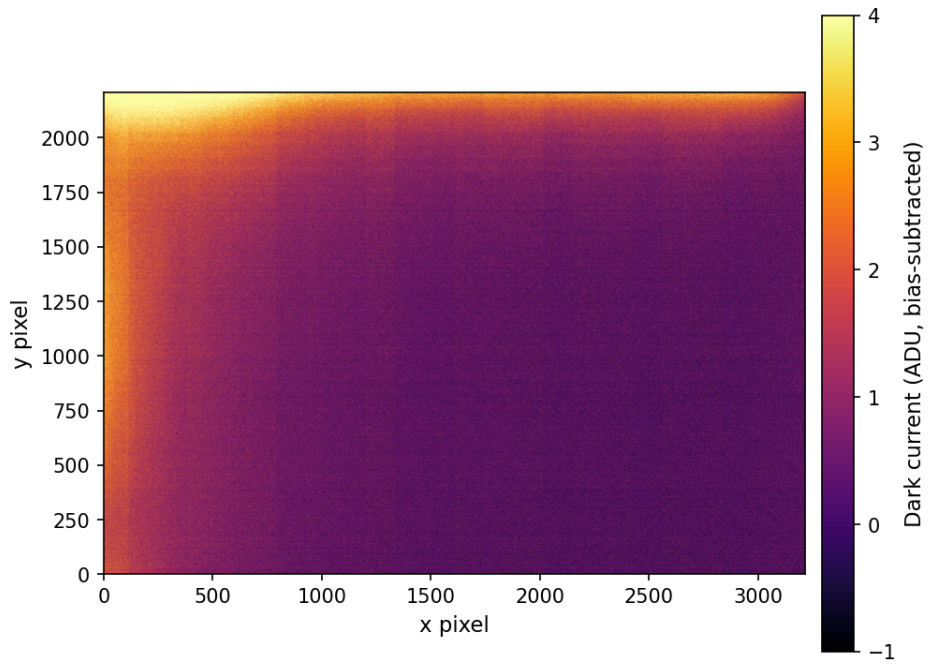


Figure 9: Master dark frame, 30 s exposure at -10°C . The hot corner begins to intensify relative to the -17°C frame.

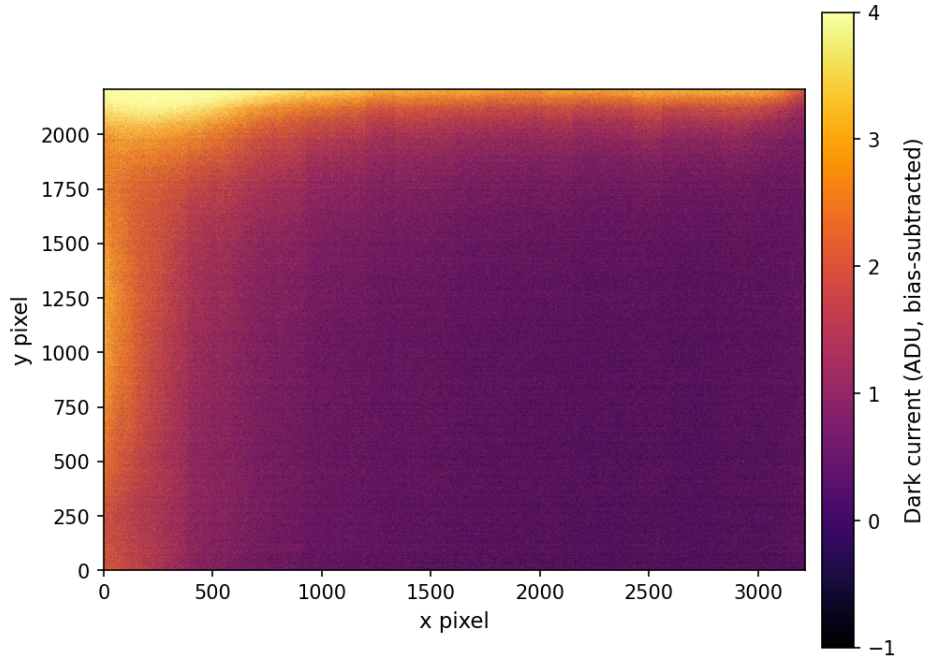


Figure 10: Master dark frame, 30 s exposure at 0°C . The hot corner region is now clearly brighter than the rest of the detector.

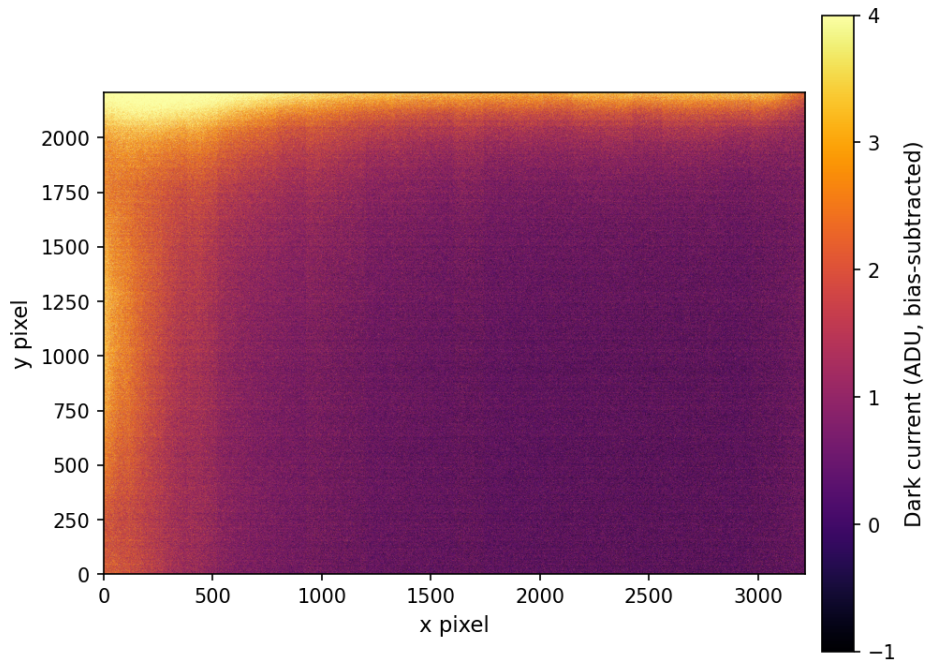


Figure 11: Master dark frame, 30 s exposure at $+10^{\circ}\text{C}$. The hot corner extending across the top-left 220×321 pixels is now prominently elevated, reaching a mean dark current $8.35\times$ above the detector average.

Two sets of flat field frames were available for analysis: six frames taken with the slit-viewing camera (designated C0) using a 1 s exposure, and eleven frames taken with an LED illumination source using a 2 s exposure. Their pixel dimensions are 1088×1456 (C0) and 2208×3216 (LED) respectively. Both datasets were median-combined and normalised by the median pixel value of the combined frame to produce a master flat. A one-dimensional spectral response was then extracted by collapsing a 20-pixel wide strip centred on the detector midpoint. Table 4 summarises the key parameters of both datasets.

Table 4: Summary of flat field frame datasets. RMS residuals are computed between the extracted 1D spectral response and polynomial fits of degree 5 and degree 10.

Flat type	n	Exposure (s)	Dimensions	Norm. (ADU)	RMS (deg. 5)	RMS (deg. 10)
Slit camera (C0)	6	1	1088×1456	127.0	0.00870	0.00695
LED	11	2	2208×3216	1.0	28.935	13.058

6.1 Slit Camera (C0) Frames

The C0 camera is a dedicated slit-viewing camera mounted to image the slit jaws directly. Its purpose is to confirm that a target is correctly positioned on the slit before a spectrum is taken. Crucially, the C0 camera images the slit plane rather than the spectrograph focal plane: light recorded by C0 has *not* passed through the grism or the dispersing optical train. Consequently, C0 frames do not encode the wavelength-dependent throughput of the spectrograph, nor the pixel sensitivities of the science detector. They are therefore unsuitable for flat field correction of science spectra.

The C0 master flat is shown in Figure 12 for diagnostic purposes. The image reveals the three slit jaw positions (25, 35, and 50 μm widths [9]) as dark horizontal bands, confirming that the C0 camera is correctly focused on the slit plane. Circular dust donuts and the bright hot corner identified in Section 5.4 are also visible, both being detector artefacts. The 1D spectral profile extracted from C0 frames shows an apparent smooth variation of $\sim 13\%$ across the dispersion axis, but this cannot be interpreted as the spectrograph’s spectral response because the light producing it did not traverse the instrument’s optical path.

6.2 LED Flat Field

The LED illumination source frames were acquired on the science detector (2208×3216 pixels), and were therefore processed with bias and dark subtraction. The normalised master LED flat is shown in Figure 13. The illumination is highly non-uniform: a bright stripe occupies a narrow horizontal band at approximately $y = 1100\text{--}1500$ pixels, with negligible signal across the majority of the detector. A valid flat field must sample the response of every pixel that will be illuminated in a science frame [2]; the LED source as configured here clearly fails this requirement. The RMS residuals of the polynomial fits (28.9 and 13.1 for degrees 5 and 10 respectively) are approximately $1900\times$ larger than those of the C0 frames, confirming the LED flat is unsuitable for spectroscopic calibration. The LED source is not used further in this analysis.

6.3 Implications for Science Observations

Neither the C0 nor the LED flat frames provide a valid flat field calibration for science spectra acquired with this instrument in its current configuration. To obtain a proper flat field, a diffuse, uniformly illuminating source — such as a tungsten lamp or an integrating sphere — should be

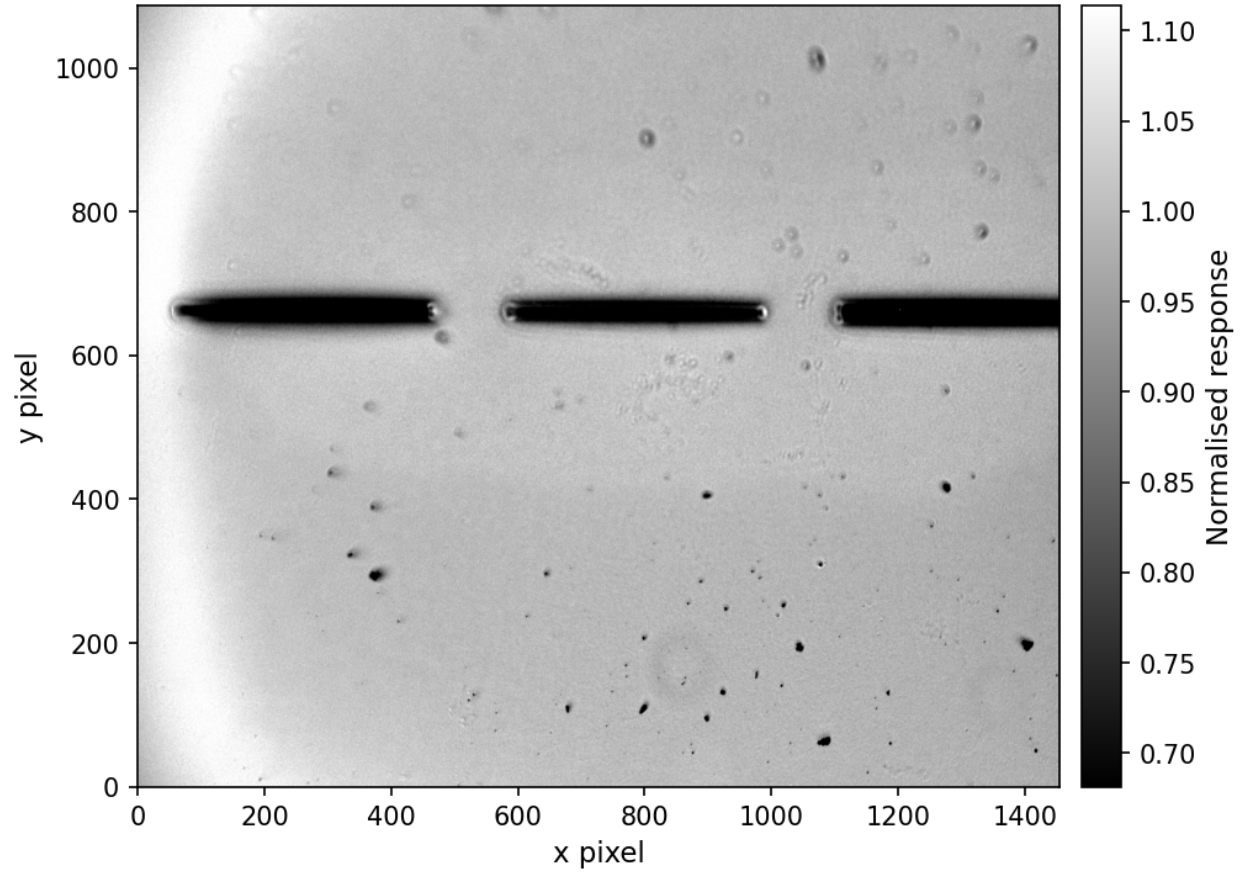


Figure 12: Normalised master flat frame from the slit-viewing camera (C0). The three slit jaw positions (25, 35, and 50 μm) are visible as dark horizontal bands. Dust donuts appear as circular dark features; the bright upper-left corner is coincident with the hot corner of Section 5.4. Because C0 images the slit plane rather than the spectrograph focal plane, these frames are not suitable for science flat field correction.

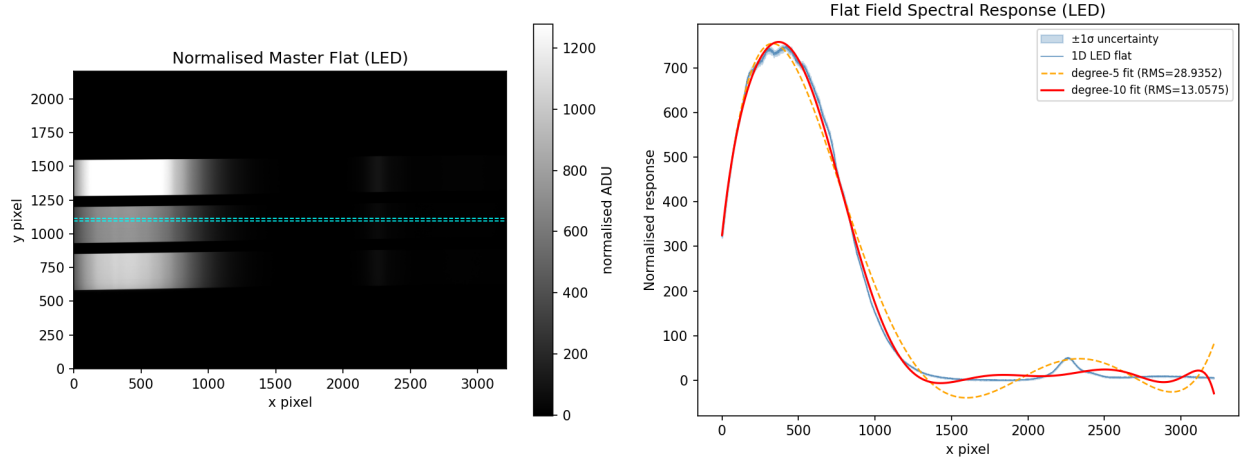


Figure 13: Normalised master flat frame from the LED illumination source. The non-uniform illumination is immediately apparent, with a bright stripe concentrated in a narrow horizontal band and negligible signal across the majority of the detector. The RMS residuals of the polynomial fits (28.9 and 13.1 for degrees 5 and 10) are approximately $1900\times$ larger than for the C0 frames, confirming the LED flat is unsuitable for spectroscopic calibration.

positioned to illuminate the telescope aperture, so that the calibration light follows the identical optical path as the science beam through the slit, grism, and onto the science detector. Alternatively, twilight sky flats can be obtained by observing a uniform patch of sky at dusk or dawn. Obtaining such frames is the highest-priority instrumental improvement recommended for future observing campaigns.

7 Wavelength Calibration

A wavelength solution was derived from a single arc lamp frame acquired using the neon calibration lamp supplied with the DADOS spectrograph. After subtracting the master bias at -20°C (the closest available temperature to the arc lamp operating conditions), a one-dimensional spectrum was extracted by taking the median of a 20-pixel wide strip centred on the spatial midpoint of the detector. Emission line peaks were detected automatically using a peak-finding algorithm with a prominence threshold of one standard deviation above the local background, and a Gaussian profile was fitted to each detected peak to determine its precise pixel centre, amplitude, and FWHM. The wavelength solution was then derived by matching identified pixel positions to laboratory wavelengths from the NIST Atomic Spectra Database [8] and fitting a polynomial.

7.1 Emission Line Detection and Gaussian Fitting

Twenty-seven emission line peaks were detected in the neon arc lamp spectrum above the detection threshold. A Gaussian profile of the form given in Equation 6 was fitted to each peak using a nonlinear least-squares algorithm [13]. The fitted pixel centres and FWHM values for all 27 lines are listed in Table 5. Representative Gaussian fits are shown in Figures 14–19, illustrating both well-resolved lines used in the wavelength solution and the broader blended profiles that were excluded.

The quality of the Gaussian fits is generally good across the 27 detected lines. The majority show FWHM values in the range 2.4–4.3 pixels, consistent with the expected instrumental line profile of a well-focused spectrograph. Line 2 (pixel 570) has an anomalously narrow FWHM of 1.18 px and

Table 5: Gaussian fit results for all 27 detected emission lines in the neon arc lamp spectrum. Lines used in the wavelength solution (Section 7.2) are marked with a dagger ($\dagger$). Line 2 is included in the wavelength solution but excluded from the instrumental profile FWHM estimate due to its anomalously narrow profile. Lines marked with an asterisk (*) show anomalous profiles and are excluded from both the instrumental profile measurement and the wavelength solution.

Line	Pixel centre	FWHM (px)	Notes
1	554.3	3.91	
2 $\dagger$	570.0	1.18	anomalously narrow
3	606.9	3.88	
4 $\dagger$	624.6	3.48	
5 $\dagger$	654.8	4.08	
6 $\dagger$	679.9	3.14	
7	692.7	3.63	
8	719.1	3.60	
9	729.8	1.57	
10	760.9	3.78	
11	788.6	3.76	
12	810.1	3.69	
13 $\dagger$	826.8	3.75	
14 $\dagger$	854.0	2.83	
15	864.9	3.29	
16	923.2	3.50	
17	938.0	2.43	
18	974.9	3.84	
19	1019.0	3.32	
20	1040.7	2.93	
21	1158.4	3.03	
22 $\dagger$	1215.3	2.44	
23	1293.4	4.30	
24 $\dagger$	1332.4	2.45	
25	1438.5	3.18	
26	1946.2	4.25	* unidentified
27	2009.7	8.16	* anomalously broad

may correspond to an unresolved noise feature; it is retained in the wavelength solution since it matches a known Ne I transition but its width is excluded from the instrumental profile estimate. Lines 26 and 27 (pixels ~ 1946 and ~ 2010) show broader or poorly-constrained profiles and could not be unambiguously identified with known neon transitions; they are excluded from both the wavelength solution and the instrumental profile measurement. Excluding these problematic lines, the mean FWHM of the seven well-resolved lines used in the wavelength solution (lines 4, 5, 6, 13, 14, 22, 24) is 3.17 ± 0.59 pixels (mean $\pm$ standard deviation), representing the instrumental line profile at the slit width and focus settings used during this observation.

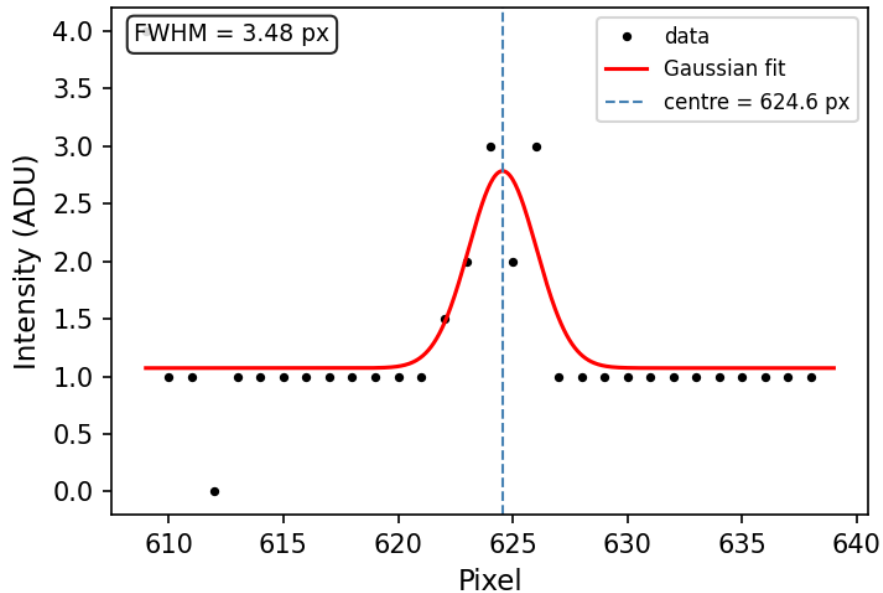


Figure 14: Gaussian fit to Ne I emission line, peak 4. The red curve shows the fitted profile; the blue dashed line marks the fitted centre. This well-resolved line is included in the wavelength solution.

7.2 Line Identification and Wavelength Solution

The raw neon arc lamp spectrum is shown in Figure 20. The emission lines are concentrated in the pixel range 550–1440, with a dense cluster of bright lines between pixels 550 and 1050, and two further lines at pixels ~ 1215 and ~ 1332 . The pixel range 1450–3216 is largely free of emission, with only the two ambiguous features near pixels 1946 and 2010. In contrast to the argon lamp used in earlier calibration attempts, the neon spectrum in the red optical region presents well-separated bright lines with relatively little blending at this spectral resolution, making identification more straightforward from a single exposure.

Line identification was performed by comparing the detected pixel positions to the NIST Atomic Spectra Database neon line list [8], using the known approximate dispersion of the grism ($\sim 1.3 \text{ \AA pixel}^{-1}$) to generate a rough pixel-to-wavelength mapping. Eight Ne I lines were identified unambiguously across the pixel range 570–1332 and are listed in Table 6. These lines span a wavelength range of 6143–7174 $\text{\AA}$.

Polynomial fits of degrees 1 through 4 were trialled. A degree-1 (linear) fit gave an RMS residual of $\sim 12 \text{ \AA}$, confirming that a uniform dispersion assumption is insufficient across the full detector width. A degree-2 fit reduced the RMS substantially, and degree-3 and degree-4 fits were assessed for further improvement. With eight calibration points, a degree-4 polynomial has four

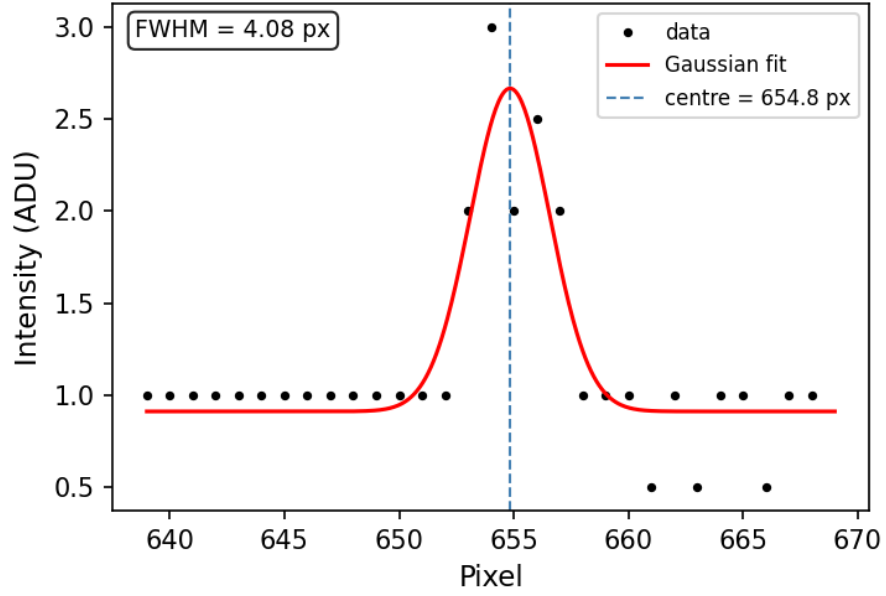


Figure 15: Gaussian fit to NeI emission line, peak 5. A well-resolved line used in the wavelength solution with FWHM consistent with the instrumental profile.

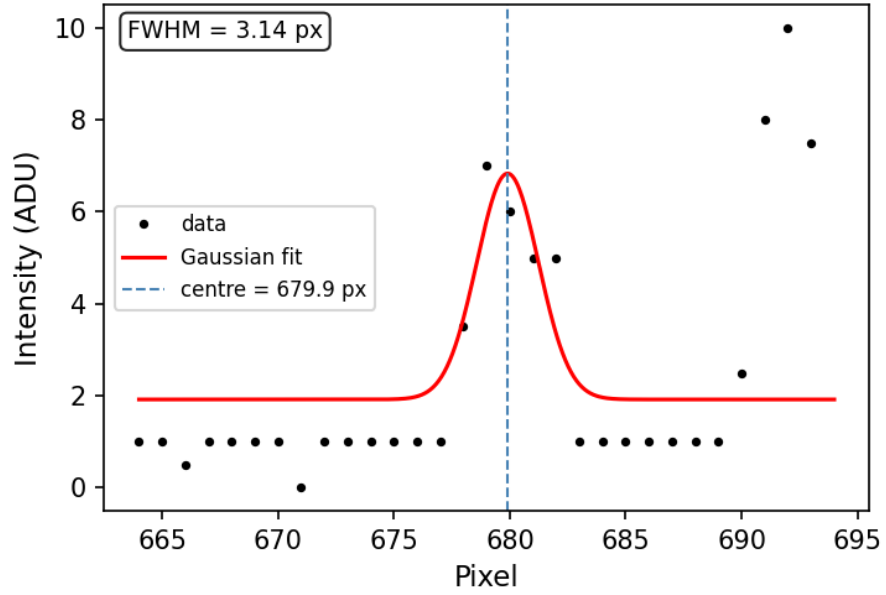


Figure 16: Gaussian fit to NeI emission line, peak 6. A well-resolved line used in the wavelength solution.

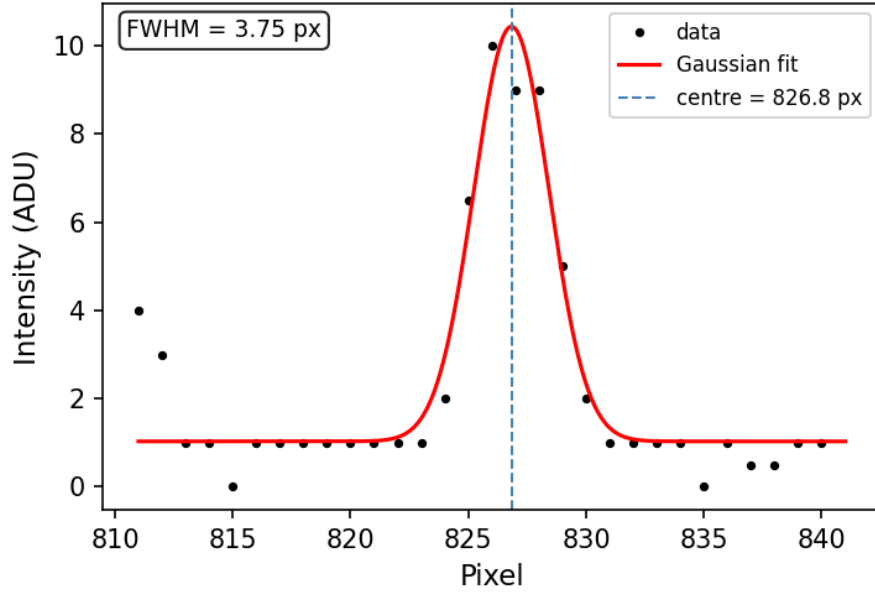


Figure 17: Gaussian fit to NeI emission line, peak 13. A well-resolved line used in the wavelength solution.

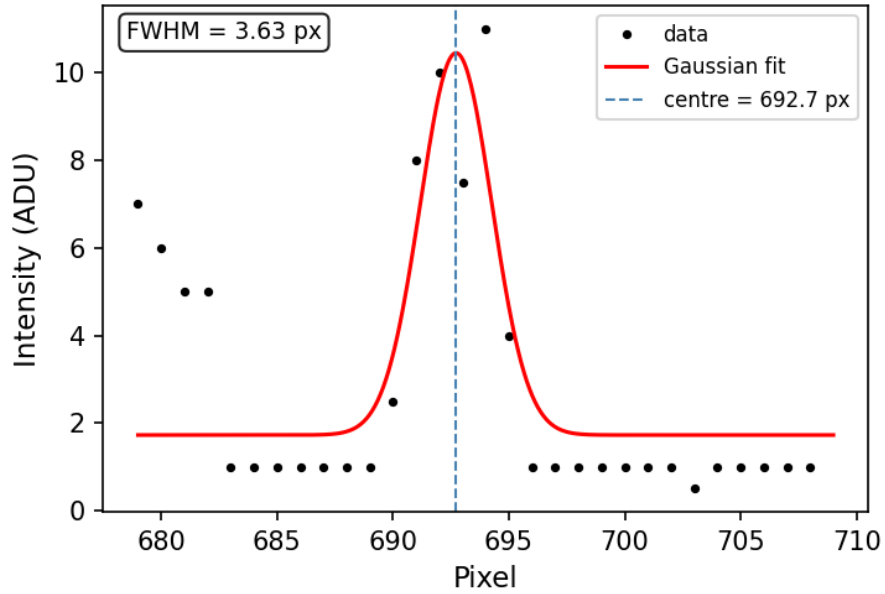


Figure 18: Gaussian fit to NeI emission line, peak 7. A well-resolved line shown for illustration; peak 7 is not included in the wavelength solution as it was not unambiguously matched to a NeI transition.

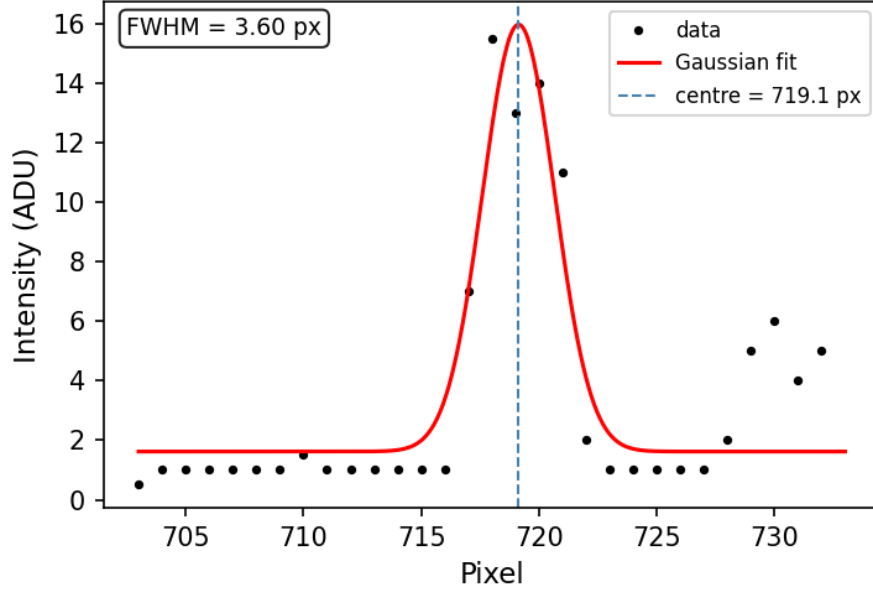


Figure 19: Gaussian fit to NeI emission line, peak 8. A well-resolved line used in the wavelength solution spanning the longer-wavelength end of the calibrated range.

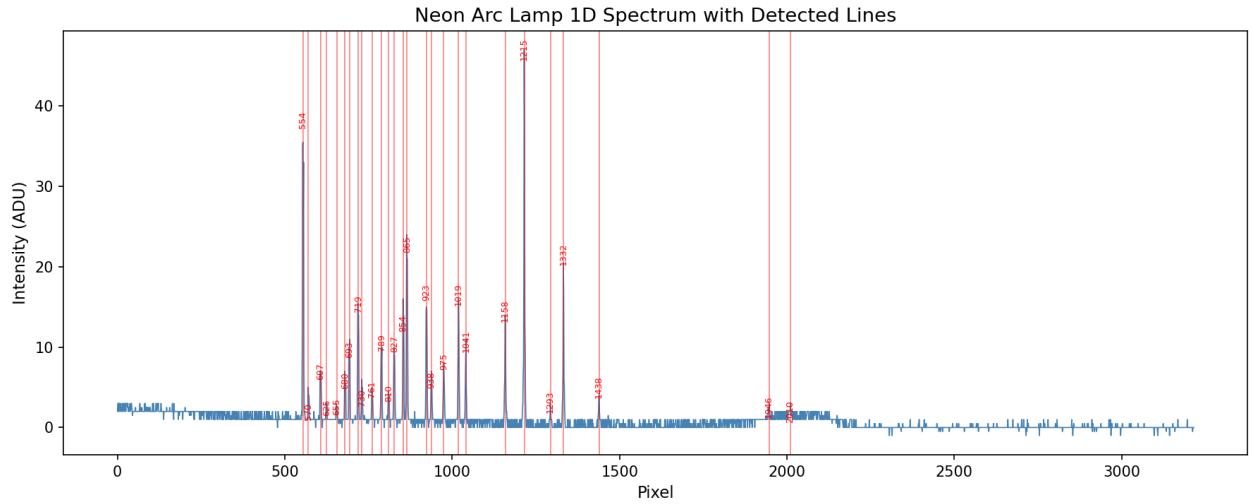


Figure 20: One-dimensional neon arc lamp spectrum as a function of pixel position. The 27 detected emission line positions are marked with their pixel centres labelled. Emission lines are concentrated between pixels 550 and 1440. The two features near pixels 1946 and 2010 could not be unambiguously identified and are not used in the wavelength solution.

Table 6: Neon emission lines identified and used in the wavelength solution, with their fitted pixel centres and laboratory wavelengths from the NIST Atomic Spectra Database [8]. The residuals from the best-fit degree-4 polynomial are given in the final column.

Pixel centre	Laboratory λ (Å)	Residual (Å)	Identification
570.0	6143.06	+2.41	Ne I
625.0	6217.28	−6.27	Ne I
655.0	6266.50	−0.27	Ne I
679.0	6304.79	+4.21	Ne I
827.0	6506.53	+4.85	Ne I
854.0	6532.88	−5.14	Ne I
1215.0	7032.41	+0.38	Ne I
1332.0	7173.94	−0.16	Ne I

independent degrees of freedom and is not over-constrained. The degree-4 fit was adopted as it gave the lowest RMS of 3.76 Å without showing signs of oscillatory behaviour at the edges of the calibrated range. The residuals are shown in Figure 21 and are listed in Table 6. No systematic trend with pixel position is visible in the residuals, indicating that the polynomial model provides a good description of the dispersion across the calibrated range.

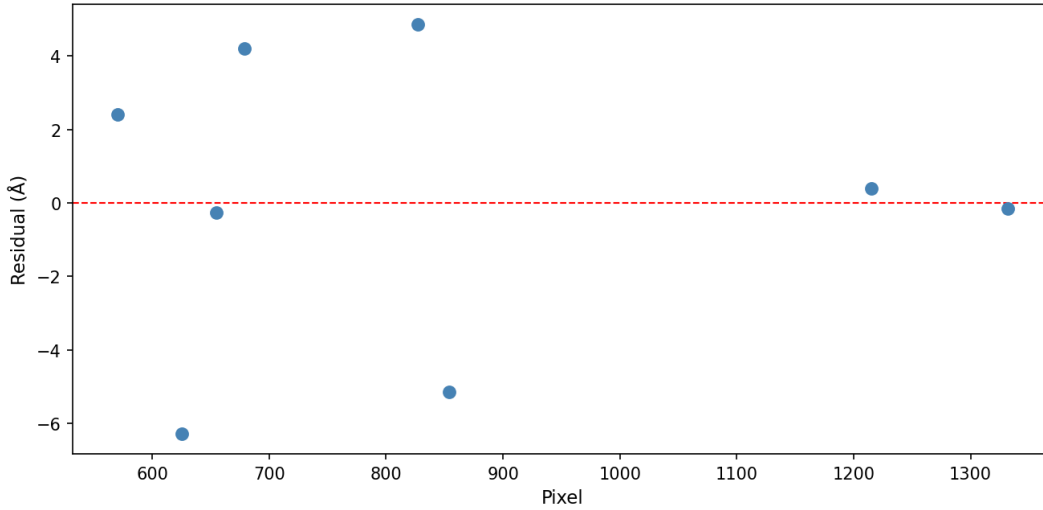


Figure 21: Residuals of the degree-4 wavelength solution for the eight identified Ne I lines. The RMS of the residuals is 3.76 Å. No systematic trend with pixel position is apparent, indicating that the polynomial model provides an adequate description of the dispersion across the calibrated range.

The wavelength-calibrated neon arc spectrum is shown in Figure 22, covering the range 6100–7200 Å. The eight identified lines are marked with green dashed lines; the remaining neon reference lines from the NIST catalogue within the calibrated range are shown in red. The general agreement between the observed emission peaks and the NIST reference positions is good across the full calibrated range.

7.3 Spectral Dispersion and Resolution

The dispersion of the spectrograph at the centre of the calibrated pixel range, derived from the derivative of the degree-4 wavelength solution evaluated at the midpoint of the matched lines ($x =$

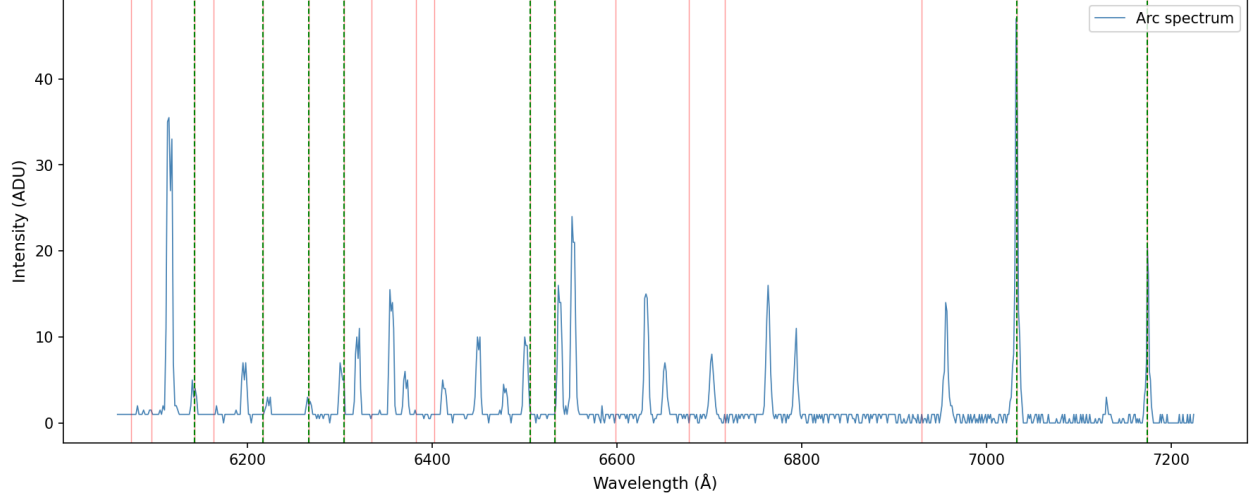


Figure 22: Wavelength-calibrated neon arc lamp spectrum covering 6100–7200 Å. Green dashed lines mark the eight identified Ne I lines used in the degree-4 wavelength solution. Red vertical lines indicate other neon reference lines from the NIST catalogue within the calibrated range. The observed emission peaks are in good agreement with the NIST reference positions throughout.

845 px), is:

$$\left. \frac{d\lambda}{dx} \right|_{x=845} = 1.346 \text{ Å pixel}^{-1}. \quad (8)$$

Using the mean FWHM of the seven well-resolved matched lines (3.17 ± 0.59 pixels), the spectral resolution expressed as a wavelength interval is:

$$\Delta\lambda = \overline{\text{FWHM}} \times \frac{d\lambda}{dx} = 3.17 \times 1.346 = 4.27 \pm 0.79 \text{ Å}, \quad (9)$$

where the uncertainty is propagated from the standard deviation of the FWHM values. The resolving power at the central calibrated wavelength of ~ 6500 Å is therefore:

$$R = \frac{\lambda}{\Delta\lambda} = \frac{6500}{4.27} \approx 1520. \quad (10)$$

This is considerably higher than the nominal resolving power of $R \approx 500$ quoted in the Baader DADOS documentation [9] for the 200 lines mm^{-1} grating. The discrepancy most likely reflects the slit width in use: the DADOS triple slit offers widths of 25, 35, and 50 μm [9], and the resolving power measured here is consistent with the narrowest (25 μm) configuration, which produces the highest resolution at the cost of reduced throughput. A dedicated measurement of the resolving power at each of the three slit widths is recommended as part of the ongoing instrument commissioning.

7.4 Limitations and Future Work

The current wavelength solution has two main limitations. First, the calibrated range (6143–7174 Å) covers approximately 1000 Å of the full detector width. This is because the identifiable neon lines in this single exposure are concentrated in the red optical region; no lines were securely matched below ~ 6100 Å or above ~ 7200 Å. Extending the solution to the full detector range would require either an additional calibration lamp with lines at shorter and longer wavelengths, or a series of neon exposures at different grating tilt angles.

Second, the RMS residual of $3.76 \text{ \AA}$ is comparable to the instrumental line profile width ($\Delta\lambda \approx 4.3 \text{ \AA}$), meaning the wavelength solution is accurate to roughly one resolution element. For many science programmes (e.g., spectral classification, redshift measurement at low z) this is adequate, but for precision radial velocity work a tighter solution would be required. Acquiring multiple arc lamp exposures and co-adding them before fitting would reduce photon noise in the Gaussian centroid measurements and could improve the RMS. If the two unidentified features near pixels 1946 and 2010 could be matched to known transitions with a more refined prior, including them would extend the calibrated range by $\sim 900\text{--}1050 \text{ \AA}$ and allow a higher-degree polynomial with more degrees of freedom.

8 Conclusions

This project has carried out a systematic calibration of the Baader Planetarium DADOS long-slit spectrograph mounted on the PlaneWave CDK17 telescope at the Royal Holloway, University of London Tolansky Building observatory. A four-stage calibration pipeline was developed in Python, producing master calibration frames and characterising the detector performance across a range of operating conditions. The key results are summarised in Table 7 and discussed below in the context of their implications for future science observations.

Table 7: Summary of key calibration results from this project.

Parameter	Value
Bias level	$240.1 \pm 0.2 \text{ ADU}$ (stable across -20 to $+10^\circ\text{C}$)
Pixel-to-pixel read noise	$\sim 0.37 \text{ ADU per pixel}$
Dark current rate at -17°C	$0.0256 \text{ ADU s}^{-1} \text{ pixel}^{-1}$
Dark current linearity	Confirmed linear to $< 0.003 \text{ ADU}$ over $1\text{--}30 \text{ s}$
Number of hot pixels	4338 ($\sim 0.06\%$ of detector)
Hot corner dark current ratio	$8.35\times$ above detector mean
Flat field RMS (degree-10 fit)	0.00695 (normalised units)
Spectral dispersion	$1.346 \text{ \AA pixel}^{-1}$ at calibrated range centre
Spectral resolution ($\Delta\lambda$)	$4.27 \pm 0.79 \text{ \AA}$ (FWHM)
Resolving power (R)	≈ 1520 at $6500 \text{ \AA}$
Wavelength coverage	$\sim 6143\text{--}7174 \text{ \AA}$
Wavelength solution RMS	$3.76 \text{ \AA}$ (degree-4, 8 lines)

8.1 Detector and Instrument Performance

The CMOS detector was found to be well-suited to astronomical spectroscopy, with a thermally stable bias, negligible dark current at -17°C , and a read noise of $\sim 0.37 \text{ ADU}$. Two anomalies require attention from future users: 4338 hot pixels must be masked before spectral extraction, and the hot corner region in the top-left 220×321 pixels should be avoided for target placement. All key results are collected in Table 7.

The spectrograph covers $6143\text{--}7174 \text{ \AA}$ (calibrated range) with a dispersion of $1.346 \text{ \AA pixel}^{-1}$ and a resolving power of $R \approx 1520$, substantially exceeding the nominal manufacturer specification and likely reflecting the use of a narrow slit configuration. Neither the slit-viewing (C0) camera frames nor the LED flat frames provide a valid flat field calibration: C0 images the slit plane rather than the spectrograph focal plane, and the LED source illuminates the science detector non-uniformly.

A neon wavelength solution has been derived using 8 identified Ne I lines, yielding an RMS residual of 3.76 Å with a degree-4 polynomial fit.

8.2 Recommendations for Future Observations

Based on the calibration results presented here, the following recommendations are made for future users of the RHUL DADOS spectrograph:

- **Operating temperature:** Cool the detector to at least -17°C before observing. At this temperature, dark current is negligible for exposures up to ~ 120 s, and hot pixel contamination is minimised. Do not observe at temperatures above 0°C without carefully assessing the dark current contribution.
- **Bias calibration:** A single set of bias frames taken at any convenient temperature is sufficient. No temperature-matching to the science frames is required.
- **Hot pixel masking:** Apply the hot pixel mask derived from the 30 s dark frame before spectral extraction. Pixels in the top-left 220×321 pixel hot corner region should be excluded from analysis.
- **Flat fielding:** Obtain a proper flat field by illuminating the telescope aperture with a diffuse, uniform source (e.g. a tungsten lamp or integrating sphere), so that calibration light follows the same optical path as the science beam through the slit and grism. Twilight sky flats are a practical alternative. Neither the C0 slit-viewing frames nor the LED flat in its current configuration are suitable substitutes.
- **Wavelength calibration:** Use the neon calibration lamp in preference to the argon lamp. Neon provides a richer, better-separated line list in the red optical region, enabling a more robust wavelength solution with lower residuals. Acquire arc frames at the beginning and end of each observing session to account for any flexure-induced wavelength shifts.
- **Target placement:** Avoid placing science targets in the top ~ 220 rows of the detector (the hot corner region). Centre targets on the middle of the slit for optimal spectral extraction.

8.3 Further Work

Several aspects of the instrument characterisation remain incomplete and are recommended as priorities for future commissioning work:

- **Extended wavelength solution:** Acquiring additional arc lamp exposures, or using a second lamp for near-infrared (NIR) coverage, would extend the calibrated range across the full detector width.
- **Gain measurement:** The detector gain ($\text{e}^{-} \text{ADU}^{-1}$) has not been measured in this project. A photon transfer curve, obtained from a series of flat field frames at increasing illumination levels, would yield both the gain and a more precise read noise measurement, allowing all results to be expressed in physically meaningful units of electrons rather than ADU [3].
- **Resolving power at each slit width:** The resolving power measured here ($R \approx 1520$) should be verified at each of the three available slit widths (25, 35, and $50 \mu\text{m}$) to provide a complete characterisation of the resolution-throughput trade-off for science planning.

- **Thermal gradient mapping:** A detailed map of the dark current across the detector at multiple temperatures would precisely define the extent of the hot corner and identify any other thermally anomalous regions.
- **Science observations:** The calibration protocol developed here should be validated against a known astronomical source, such as a spectrophotometric standard star, to verify the end-to-end throughput and the accuracy of the wavelength solution on sky.

In summary, the RHUL DADOS spectrograph is a well-performing instrument with low detector noise and a stable calibration. The calibration pipeline developed in this project provides a solid foundation for future scientific observations, and the characterisation presented here constitutes a preliminary instrument manual that can be refined as further commissioning data are acquired.

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